034IN -FONDAMENTI DI AUTOMATICA - FUNDAMENTALS OF AUTOMATIC CONTROL

FULL WRITTEN EXAMINATION - A.Y. 2023/2024

September 2, 2024

PART 1 - Problem 2

Tag List: `[STANDARD ASSESSMENT] [PARTIAL CREDIT] [SYMBOLIC MATH TOOLBOX] [CONTROL TOOLBOX]`

Consider the LTI dynamic system described by the following differential equations

$$\begin{cases} \dfrac{\mathrm{d}^2\vartheta(t)}{\mathrm{d}t^2} &= k_1\left[u(t) - m_1\dfrac{\mathrm{d}\vartheta(t)}{\mathrm{d}t} - m_2\,k_2\,\vartheta(t) - m_3\dfrac{\mathrm{d}z(t)}{\mathrm{d}t}\right] \\ \dfrac{\mathrm{d}^2 z(t)}{\mathrm{d}t^2} &= k_2\vartheta(t) \end{cases}$$

where

$$k_1 = 4.815\,, \quad k_2 = 1.623$$
$$m_1\,,\, m_2\,,\, m_3 \in \mathbb{R}$$

and $u(t)$ is a proper causal input signal.

Moreover, assume the following two outputs for the system

$$\begin{cases} y_1(t) &= \vartheta(t) \\ y_2(t) &= \dfrac{\mathrm{d}z(t)}{\mathrm{d}t} \end{cases}$$

Assignment

Your code shall:

- Declare the parameters $m_1, m_2,$ and m_3 as the **symbolic variables `m1`**, **`m2`**, and **`m3`**. Moreover, declare the parameters k_1 and k_2 as the symbolic variables **`k1`** and **`k2`**. Finally, declare as symbolic variables **`k1val`** and **`k2val`** the values to be assigned to k_1 and k_2. **Do not assign any value** to **`k1`** and **`k2`** for the moment.
- Determine a state-space description for the LTI system. Assume the state variables as follows:

$$\begin{cases} x_1(t) &= \vartheta(t) \\ x_2(t) &= \dfrac{\mathrm{d}\vartheta(t)}{\mathrm{d}t} \\ x_3(t) &= \dfrac{\mathrm{d}z(t)}{\mathrm{d}t} \end{cases}$$

- Assign the matrices A, B, C, D of the state equations respectively to the **symbolic variables `symA`**, **`symB`**, **`symC`**, and **`symD`**.
- Calculate the characteristic polynomial of the LTI system as a symbolic function of the complex symbolic variable s and save the resulting expression in the symbolic function **`pA(s)`**.
- **Note**: `pA(s)` depends on the symbolic variables **`m1`**, **`m2`**, and **`m3`** and the symbolic parameters **`k1`** and **`k2`**.
- Now **assign** the values **`k1val`** and **`k2val`** respectively to the parameters **`k1`** and **`k2`** into the polynomial **`pA(s)`**, using the MATLAB command **`subs`**.
- Determine proper values for the parameters `m1`, `m2`, and `m3`, such that the LTI eigenvalues are $\lambda_1 = \lambda_2 = \lambda_3 = -3$, i.e. the characteristic polynomial `pA(s)` becomes $(s+3)^3$. Store the just obtained parameter values in the **symbolic variables `M1p`**, **`M2p`**, and **`M3p`**.
- Substitute the values **`M1p`**, **`M2p`**, and **`M3p`** toghether with the values **`k1val`** and **`k2val`** in the matrix **`symA`** and verify that the eigenvalues are effectively $\lambda_1 = \lambda_2 = \lambda_3 = -3$. Save the eigenvalues in the **symbolic variables `lambda1`**, **`lambda2`**, and **`lambda3`**.

Solution

```
clear
close all
clc
```

```
syms k1 k2
syms m1 m2 m3
syms s

k1val = sym(4.815);
k2val = sym(1.623);

symA = [     0         1        0; ...
        -m2*k1*k2   -m1*k1   -m3*k1; ...
            k2         0        0];

symB = [  0; ...
          k1; ...
          0];

symC = sym([1   0   0; ...
            0   0   1]);

symD = sym([0;0]);
```

```
pA(s) = charpoly(symA, s)
```

pA(s) = $s^3 + k_1\, m_1\, s^2 + k_1\, k_2\, m_2\, s + k_1\, k_2\, m_3$

```
pA(s) = subs(pA, [k1 k2], [k1val, k2val])
```

pA(s) =

$$s^3 + \frac{963\, m_1\, s^2}{200} + \frac{1562949\, m_2\, s}{200000} + \frac{1562949\, m_3}{200000}$$

```
targetP = expand((s+3)^3)
```

targetP = $s^3 + 9\, s^2 + 27\, s + 27$

```
% coeffsA = coeffs(pA, s) % NB <<--- coeffsA is a symfunction  !!!!
Apoly = charpoly(symA,s);
coeffsA = subs(coeffs(Apoly, s), [k1 k2], [k1val, k2val])
```

coeffsA =

$$\begin{pmatrix} \frac{1562949\, m_3}{200000} & \frac{1562949\, m_2}{200000} & \frac{963\, m_1}{200} & 1 \end{pmatrix}$$

```
coeffsT = coeffs(targetP, s)
```

coeffsT = $(27 \;\; 27 \;\; 9 \;\; 1)$

```
EQ0 = coeffsA(1) == coeffsT(1);
EQ1 = coeffsA(2) == coeffsT(2);
EQ2 = coeffsA(3) == coeffsT(3);

EQ_M = [EQ2; EQ1; EQ0]
```

EQ_M =

$$\begin{pmatrix} \frac{963\, m_1}{200} = 9 \\ \frac{1562949\, m_2}{200000} = 27 \\ \frac{1562949\, m_3}{200000} = 27 \end{pmatrix}$$

```
Msol = solve(EQ_M, [m1 m2 m3])
```

```
Msol = struct with fields:
    m1: 200/107
    m2: 200000/57887
    m3: 200000/57887
```

```
M1p = Msol.m1
```

M1p =

$$\frac{200}{107}$$

```
M2p = Msol.m2
```

M2p =

$$\frac{200000}{57887}$$

```
M3p = Msol.m3
```

M3p =

$$\frac{200000}{57887}$$

```
% --- alternative approach ---
% by visual comparison of the polynomials, write the linear algebra problem
%                    matM * unknownV = matCoeff
% where:
%
% matM = [k1val         0               0; ...
%           0       k1val*k2val           0; ...
%           0           0           k1val*k2val]
%
% matCoeff = sym([9; 27; 27])
%
% Msol = linsolve(matM, matCoeff)
%
% M1p = Msol(1)
% M2p = Msol(2)
% M3p = Msol(3)

symA = subs(symA, [m1 m2 m3 k1 k2], [M1p M2p M3p k1val k2val])
```

symA =

$$\begin{pmatrix} 0 & 1 & 0 \\ -27 & -9 & -\frac{9000}{541} \\ \frac{1623}{1000} & 0 & 0 \end{pmatrix}$$

```
lambdaA = eig(symA)
```

lambdaA =

$$\begin{pmatrix} -3 \\ -3 \\ -3 \end{pmatrix}$$

```
lambda1 = lambdaA(1)
```

lambda1 = -3

```
lambda2 = lambdaA(2)
```

lambda2 = -3

```
lambda3 = lambdaA(3)
```

lambda3 = -3